

CO-2 Analytical problem Solving

1)

$$\begin{array}{r}
 11010011 \\
 1001 \overline{) 110001011000} \\
 \underline{1010} \\
 1001 \downarrow \\
 \underline{0111} \\
 0000 \downarrow \\
 \underline{1110} \\
 1001 \downarrow \\
 \underline{1111} \\
 0000 \downarrow \\
 \underline{1111} \leftarrow 001 \\
 0000 \downarrow \\
 \underline{1111} \leftarrow 1001 \\
 1001 \downarrow \\
 \underline{1100} \\
 1001 \downarrow \\
 \underline{1010} \\
 1001 \downarrow \\
 \underline{0110} \\
 0000 \downarrow \\
 \underline{110} \leftarrow \text{remainder}
 \end{array}$$

Transmitted sequence - 110001011110

2. $G(x) = x^3 + x + 1$

binary pattern $\rightarrow 1011$ appended message $\rightarrow 11000000$

$$\begin{array}{r}
 1110 \\
 1011 \overline{) 11000000} \\
 \underline{1011} \\
 1110 \\
 \underline{1011} \\
 1010 \\
 \underline{1011} \\
 0010 \\
 \underline{0011} \\
 0010 \\
 0000 \\
 \hline
 0010 \\
 0000 \\
 \hline
 100 \leftarrow \text{remainder}
 \end{array}$$

Transmitted Codeword $\rightarrow 11000100$

3.
$$\begin{array}{cccccccccccc}
 P_{12} & P_{11} & P_{10} & P_9 & P_8 & P_7 & P_6 & P_5 & P_4 & P_3 & P_2 & P_1 \\
 d_8 & d_7 & d_6 & d_5 & C_8 & d_4 & d_3 & d_2 & C_4 & d_1 & C_2 & C_1 \\
 1 & 1 & 0 & x & y & 0 & 1 & 0 & 0 & 1 & 1 & 0
 \end{array}$$

For (2) $P(2, 3, 6, 7, 10, 11)$
 $1, 1, 1, 0, 0, 1 = 0$

For (4) $P(4, 5, 6, 7, 12)$

$0, 0, 1, 0, 1 = 0$

for (8) $P(8, 9, 10, 11, 12)$

$y, x, 0, 1, 1 = 0$

$y, 0, 0 \Rightarrow y = 0$

$x = 0, y = 0$

110000100110

4. Given,

$$C_1 = 00000$$

$$C_2 = 01011$$

$$C_3 = 10101$$

$$C_4 = 11110$$

$$(C_1, C_2) = d(00000, 01011) \\ p(2, 4, 5)$$

$$(C_2, C_3) = (01011, 10101) \\ p(1, 3, 5)$$

$$(C_3, C_4) = (10101, 11110) \\ p(2, 4, 5)$$

$$(C_1, C_4) = (00000, 11110) \\ p(1, 2, 3, 4)$$

$$(C_2, C_4) = (01011, 11110) \\ p(1, 3, 5)$$